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Contraceptive vaginal ring use for women has less adverse metabolic effects than an oral contraceptive☆

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Abstract

Background: This study compared metabolic, hormonal and lipid profiles before and during use of a contraceptive vaginal ring (RING) releasing 15 mcg ethinyl estradiol (EE) and 120 mcg etonogestrel per day (NuvaRing®, Organon USA Inc., Roseland, NJ) versus a low-dose oral contraceptive (PILL) containing 20 mcg EE and 100 mcg levonorgestrel daily (Aviane, Barr Pharmaceuticals Inc., Pomona, NY).
Study Design: Sixty-five women were randomized to either the RING or PILL treatment for five cycles. In the pretreatment cycle (Cycle Days 2–5) and during Weeks 2 and 3 of the fifth treatment cycle, a 75-g oral glucose tolerance test (OGTT) was performed. Baseline samples were used to evaluate basal hormonal, metabolic and lipid levels.
Results: Forty-two women completed the study. Basal insulin resistance (HOMA-IR) was slightly decreased, whereas a significant reduction in the insulin sensitivity index (IS_{OGTT}) was found in women on PILL therapy compared to those in the RING group ($p<.035$). Pancreatic β-cell function was not significantly altered with either treatment.
Conclusion: The lower-dose, nonoral hormonal RING had a lesser impact on carbohydrate metabolism and greater reduction of free androgen and dehydroepiandrosterone sulfate levels than PILL treatment.
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Keywords: Metabolic change; Vaginal hormonal contraception; Insulin sensitivity; Oral contraceptive

1. Introduction

Oral contraceptives (OCs) remain the most frequently used therapy for women seeking prevention of pregnancy in the United States. Research efforts have been directed primarily toward maximizing the benefits of contraceptives while minimizing their potential risks, specifically venous thromboembolic disorders. To date, the effects of OCs on metabolic function have not been comprehensively or systematically studied. Studies evaluating the use of OCs have been performed in populations that differ in anthropometric parameters, age and genetic makeup [1]. In addition, OCs with varying doses of estrogen and dose and diverse types of progestins have been studied [2]. Despite the heterogeneity, there are numerous reports suggesting that OC therapy could worsen glucose tolerance and augment insulin resistance in both healthy women and those with polycystic ovary syndrome (PCOS) [2–12].

Insulin resistance is among the best-documented metabolic effects of progesterone itself. It is a property of many androgens and related compounds such as the progestins used in OCs [13,14]. Many studies have indicated that the progestational component of combination OCs induces an impairment of insulin sensitivity and increases insulin response after a glucose challenge in healthy users [2,3,6], whereas others report only slight changes in glucose tolerance, fasting insulin, fasting glucose and areas under glucose or insulin curves in OC users [5,7,8]. Some newer OC formulations contain less androgenic progestins such as

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desogestrel and drospirenone or lower doses of the more androgenic progestins, with a number of investigators reporting a concomitant reduction in effects on carbohydrate parameters. Gaspard et al. [15] found no deleterious effect of using OCs containing either desogestrel or drospirenone on glucose tolerance in a group of healthy women for a period of 13 months. Conversely, Petersen et al. [9] evaluated the effect of two contraceptives containing progestins with a low androgenic potency on insulin action using a frequently sampled intravenous glucose tolerance test; the users of either product showed increased concentrations of fasting plasma insulin and a decrease in insulin sensitivity. Similarly, administration of two different formulations of combined low-dose OCs was recently reported to increase insulin resistance in adolescent girls with PCOS [11].

Efforts to document the impact of OCs on carbohydrate metabolism have been hampered by the use of varying laboratory methods to assess insulin sensitivity. The euglycemic clamp is considered by many to be the gold-standard technique for estimation of insulin sensitivity. The glucose clamp method is laborious, costly and relatively invasive, requiring multiple blood sampling. Accordingly, there have been no population-based studies in healthy women using the clamp and distinguishing the effects of OC on insulin sensitivity to date. The difficulty in performing this procedure on large samples has led to the use of other simpler measures of insulin sensitivity based on a single glucose and insulin value [16]. Several diabetologists have proposed the oral glucose tolerance test (OGTT) as the better method to evaluate insulin sensitivity and secretion as well as the preferred method of diagnosing impaired glucose tolerance (IGT) and diabetes [17–19]. Although OGTTs are more difficult to perform than simple measurements of fasting glucose and insulin levels, the OGTT is a minimal risk procedure that is applicable for large-scale screening and repeated individual studies. The aim of this study was to examine the changes in carbohydrate metabolism in women before and during use of a novel form of hormonal contraception, the vaginal ring (RING), compared to a low-dose monophasic OC (PILL) employing insulin sensitivity and secretion indices derived from both fasting levels and after a glucose challenge. The avoidance of gastrointestinal metabolism with nonoral routes of administration, such as the vaginal ring, and the relatively steady-state serum concentrations that can be obtained with controlled-release formulations offer the possibility of achieving adequate contraceptive efficacy and good cycle control with lower dosing than that given with an oral formulation [20–25]. The primary objective of this trial was to determine whether vaginal hormone administration ameliorates the deterioration in carbohydrate metabolism, specifically insulin sensitivity, observed in women of reproductive age on OC medications. The secondary objectives were to examine the impact on sex steroid levels and lipid profiles.

2. Materials and methods

This was an outpatient prospective randomized, comparative study with two treatment groups over five consecutive cycles. Sixty-five healthy women aged 18–40 years seeking contraception or desiring to use contraceptive steroids for cycle control (e.g., women with irregular menses) were enrolled in the study. Patients were excluded if they had contraindications to contraceptive steroids; used drugs that interfered with carbohydrate metabolism; previously used an injectable hormonal method of contraception (within 6 months) or a hormone-medicated intrauterine device or hormonal OC (within 2 months); presented with certain conditions of relevance to vaginal ring use, including cervicitis, vaginitis or bleeding cervical erosion; any Papanicolaou (Pap) smear result other than normal, ASCUS or reactive changes; or prolapse of uterine cervix, cystocele and/or rectocele. All subjects were nonsmokers or had stopped smoking for a minimum of 3 months to participate in the study.

The Institutional Review Board of the Woman's Hospital Foundation approved the study protocol, and all subjects entering the trial signed written informed consent. After consenting, the participants were randomized using a computer-generated program to either of two treatment groups: RING (NuvaRing®, Organon USA Inc., Roseland, NJ), a contraceptive vaginal ring releasing 15 mcg ethinyl estradiol (EE) and 120 mcg etonogestrel per day, or PILL (Aviane, Barr Pharmaceuticals Inc., Pomona, NY), a low-dose OC containing 20 mcg EE and 100 mcg levonorgestrel daily.

At initial screening, all subjects provided a medical and gynecological history and documentation of a Pap result within the past 12 months. Blood pressure, height, weight and waist-to-hip girth ratio (WHR) were recorded at pretreatment. Body mass index (BMI) was calculated as weight (kg)/height $(m)^2$, and WHR was used as a measure of body fat distribution. Height was measured to the nearest centimeter using a wall-mounted stadiometer, and weight was measured by a sliding weight balance to the nearest 0.1 kg. Waist circumference was measured at the minimum circumference between the rib cage and iliac crest, and hip circumference was measured at the level of the largest circumference around the buttocks. The circumference measurements were taken in the upright position using a 15-mm-wide flexible metric tape held close to the body but not tight enough to indent the skin.

Laboratory chemistry assessment at pretreatment included a basal metabolic panel and complete blood count. A standard 75-g OGTT was performed in the morning (until 10:00 a.m.) after a 12-h overnight fast. During the OGTT, blood samples were drawn at 0 (baseline before the glucose load), 30, 60 and 120 min following glucose ingestion for the measurement of glucose and insulin. Blood chemistry, lipid profile and levels of sex hormone-binding globulin

(SHBG), dehydroepiandrosterone sulfate (DHEAS), total testosterone (T) and estradiol (E_2) were determined from the baseline sample. The free androgen index (FAI) was calculated according to the following equation: FAI= [100×T (ng/dL)/28.84×SHBG (nM)]. A urine pregnancy test was performed to confirm that the subject was not pregnant prior to insertion of the first vaginal ring or oral pill start. Hormonal contraceptive treatment was started only if this test was negative.

Enrolled subjects were assigned to receive either RING or PILL, and 6 months of medication was dispensed following pretreatment testing. All subjects used either the RING or PILL according to the recommended regimen of use. All contraception was started during the first 5 days of the cycle. Patients started on contraceptive treatment on the same day as their pretreatment assessment. All subjects were treated for five cycles of 28 days, each cycle having 3 weeks of ring or pill use followed by one ring- or pill-free week. All participants were instructed to use a barrier form of contraception (such as a condom) during the first month. Subjects received written and verbal instructions on contraceptive use. Assessments for all participants occurred between Days 2 and 5 of the first treatment cycle (prior to medication start) and during Week 2 or Week 3 (Days 8–21) of the fifth ring-/pill-consecutive cycle. All endocrine and metabolic testing performed at pretreatment was repeated at the last study visit. General medical and gynecological histories were taken only at screening.

During the entire study period, compliance to the drug treatment was documented. The occurrence of adverse events (AEs) and use of concomitant medication were recorded throughout the trial. In addition, any events that were considered directly related to ring/pill use, such as vaginal discomfort and breakthrough bleeding, were recorded in specific categories. Subjects who withdrew from the study were asked to classify their reason for withdrawal as it related to an AE, a bleeding irregularity, desire for pregnancy or "other" reasons (mainly nonmedically or compliance-to-testing related). A short compliance survey was administered at the completion of the study to patients using the RING.

2.1. Assays

For the OGTT, all blood samples were obtained by venipuncture and evaluated in the Woman's Hospital Pathology Laboratory. Glucose concentrations were analyzed in all samples before and after oral glucose challenge using the glucose oxidase method (Glucose Reagent Kit, Bayer, Newbury, UK). Baseline and posttreatment serum levels of insulin were determined in all samples in duplicate by microparticle enzyme immunoassay (Abbott AxSYM System, Abbott Laboratories, Abbott Park, IL). Circulating levels of E_2, T, DHEAS and SHBG were measured with specific chemiluminescence immunoassays (Diagnostic Products, Los Angeles, CA). Levels of E_2, T, DHEAS and SHBG were measured in the first sample of the OGTT. Levels of total cholesterol, high-density lipoprotein (HDL) and triglycerides were determined in the initial basal sample using standard enzymatic colorimetric assays on an automated clinical chemistry analyzer. Low-density lipoprotein (LDL) was calculated using the Friedewald formula. The intra- and interassay coefficients of variation were <10% for all assays performed. To avoid interassay variability, we analyzed samples from each subject together in the same assay.

2.2. Data analysis

From each OGTT, the following validated insulin indices were computed using established formulas [16–18]. Insulin sensitivity was calculated using the homeostasis model assessment (HOMA) index [16] and the composite whole-body insulin sensitivity index (IS_{OGTT}) [17]. The HOMA index was calculated from simultaneously sampled fasting insulin and glucose concentrations as follows: fasting I concentration (μU/mL)×fasting Glu concentration (mmol/L)/22. The HOMA index, which is based on fasting basal values of insulin and glucose, has previously been shown to correlate well with the euglycemic–hyperinsulinemic clamp in the assessment of insulin sensitivity in both normal and diabetic subjects [16] and may be thought of as primarily illustrating hepatic insulin sensitivity [17]. The Matsuda index of insulin sensitivity (IS_{OGTT}) is a composite measure of whole-body insulin sensitivity that includes both components of hepatic and peripheral (muscle) tissues and utilizes both fasting and stimulated (during an OGTT) values of insulin (μU/mL) and glucose (mg/dL). IS_{OGTT} is calculated as follows: IS_{OGTT}=10,000/(FPG×FSI)×(mean OGTT Glu concentration×mean OGTT I concentration), where FPG and FSI denote fasting plasma glucose and fasting serum insulin, respectively; the higher the IS_{OGTT} is, the higher is the insulin sensitivity [17]. The Matsuda index correlates well (R=.70) with insulin sensitivity measured with euglycemic insulin clamp, the gold standard for measuring insulin sensitivity and insulin secretion [17]. Insulin secretion (β-cell function) was estimated using the corrected insulin response at glucose peak (CIRgp) that is based on the dose–response relations between glucose and insulin concentrations after oral glucose loading. The CIRgp, an indicator of β-cell activity, is calculated from the OGTT data: $CIRgp=(I_{glupeak}\times 100)/(Glu_{glupeak}\times[Glu_{glupeak}-70])$; the higher the CIR_{gp} is, the greater is the insulin secretion for the same glucose stimulus [26].

2.3. Statistical analyses

All measures of insulin sensitivity, β-cell function, hormone levels and lipid concentrations were analyzed using SS/treatments×repeated measures design (measures at pretreatment and >18 weeks of treatment). Other statistical methods applied included the following (when applicable): post hoc power analysis and multiple logistic regressions to correct for confounding variables. The

primary efficacy variables were insulin sensitivity and β-cell function. Secondary efficacy variables included anthropometric, hormonal and lipid measures. All data were analyzed using SPSS 15.0 for Windows (SPSS, Inc., Chicago, IL). The safety analysis was performed with the use of summary statistics and frequency tables. All p values are two tailed; $p<.05$ was considered statistically significant. All data are expressed as means±SE.

3. Results

3.1. Subject disposition

Sixty-five women were enrolled and randomly allocated to treatment with the RING (n=34) or PILL (n=31). Fig. 1 illustrates the flow of patients through the trial including study completion rates by study assignment and early exit. Of the 65 women consented and randomly assigned,

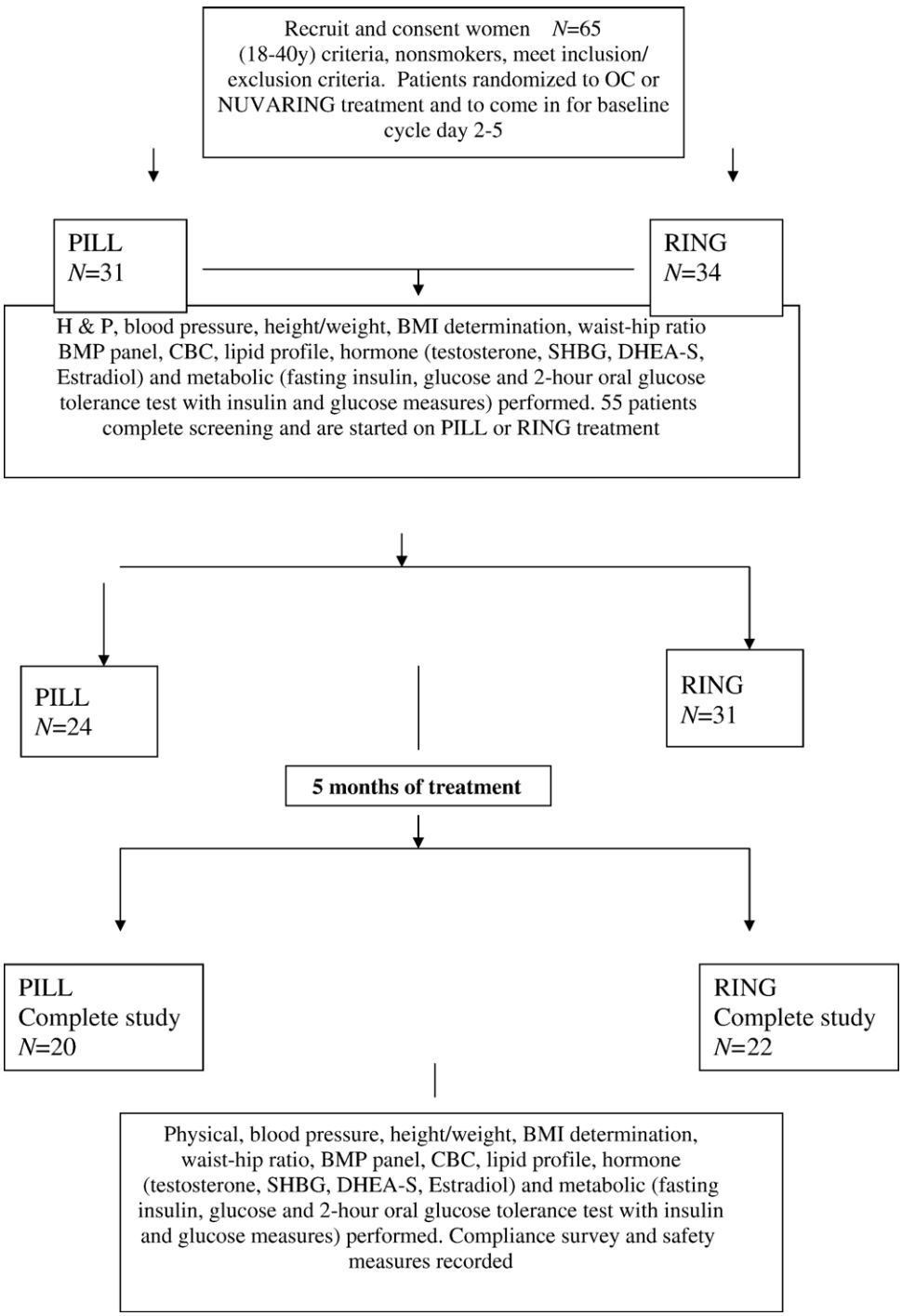

Fig. 1. Subject flowchart.

3 women never inserted a ring and 7 women never began pill intake for nonmedical reasons, making the intent-to-treat population 55 women. Of the remaining 55 randomized subjects who initiated treatment medication (RING, *n*=31; PILL, *n*=24), 13 subjects discontinued treatment prematurely (RING, *n*=9; PILL, *n*=4). Women who withdrew did so because of an inability to comply with the study medication protocol or failure to come in for a second evaluation during the fifth cycle of treatment (Fig. 1); their initial data were excluded from the data analyses. Of the study population, a total of 42 subjects (65%) used contraception and completed the 5-month study protocol (RING, *n*=22; PILL, *n*=20).

3.2. Clinical, hormonal and metabolic comparisons of study participants

The anthropometric, hormonal and biochemical results for the PILL and RING groups at baseline and during the fifth treatment month evaluation are given in Table 1. There were no significant differences between the groups with respect to mean age, BMI, WHR or hormonal and metabolic parameters. The baseline characteristics of the women in both groups were comparable (Table 1). Ethnicity was similar (80% Caucasian, 20% African American) between the two treatment groups.

Comparisons of insulin sensitivity (HOMA, IS_{OGTT}) and β-cell function (CIRgp), anthropometric parameters (BMI, WHR), hormone concentrations and lipid levels in 42 women at baseline and after 5 months of contraceptive therapy are summarized in Table 1. Physical examinations and gynecological history revealed very few clinically relevant abnormalities and no apparent differences between the treatment groups. Clinically relevant abnormalities for hematology and biochemistry values rarely occurred, and there were no significant differences between the contraceptive treatment groups in these parameters, including mean blood pressure, hemoglobin or anthropometric measures at baseline or exit. Mean BMI and WHR remained unchanged in both groups after hormonal contraceptive treatment (Table 1).

Fig. 2 illustrates changes in insulin action during contraceptive treatment. The HOMA calculation, which uses a single fasting glucose and insulin level, was moderately correlated (r=.6, p<.008) with the IS_{OGTT}, as determined from an OGTT. Fig. 2A illustrates that the insulin resistance determined with the HOMA calculation was significantly higher when the PILL pretreatment and posttreatment values were compared to each other (p<.04). Fig. 2B shows that mean insulin sensitivity measured with the IS_{OGTT} was significantly decreased after PILL treatment relative to the RING treatment (p<.035) and to pretreatment PILL measures (p<.001). No significant reduction was measured in the RING group. Two patients in the PILL group developed IGT at the time of fifth consecutive month of pill cycle assessment; no IGT was found in RING therapy patients. β-Cell function calculated from the OGTT (CIRgp) did not change consistently during the study period with either treatment (Table 1).

Circulating concentrations of both ovarian and adrenal androgens were significantly affected by hormonal contraceptive intervention. Mean T concentrations decreased while SHBG concentrations increased significantly during treatment in both groups (Table 1). The mean FAI decreased during treatment in the PILL group (p<.003) and the RING group (p<.0005), with the effect being larger in the RING group as illustrated in Fig. 3A. Fig. 3B shows that the mean blood levels of DHEAS were reduced significantly during

Table 1
Anthropometric parameters as well as metabolic and hormonal characteristics of study participants before and after 5 months of contraceptive therapy; values are expressed as means±SE

Parameter	Pill group		NuvaRing® group		p
	Pretreatment	Posttreatment	Pretreatment	Posttreatment	
Age (years)	29.5±4.7	29.9±4.8	27.9±6.7	28.4±6.6	NS
BMI (kg/m^2)	29.5±6.8	30±6.5	27±5.9	26.4±5.8	NS
WHR	0.78±0.06	0.76±0.2	0.76±0.06	0.76±0.06	NS
T (ng/dL)	43.7±6.7	29.1±3.1	50.9±6.4	26±3.0	A, <.001
SHBG (nmol/L)	50.0±6.2	75.8±12.7	52.4±5.9	163.0±12.0	A, <.001 B, <.001
Cholesterol (mg/dL)	192.6±6.9	195.5±7.0	175.2±6.6	176±6.6	NS
Triglyceride (mg/dL)	85.8±6.9	98.6±9.7	85.9±6.5	98.5±10.7	NS
HDL (mg/dL)	62.7±2.9	60.3±2.6	61.2±2.7	63.5±2.5	NS
LDL (mg/dL)	112.3±6.6	117.8±8.6	99.4±6.3	94.1±6.3	NS
HOMA	2.25±2.2	3.45±3.0	2.4±1.9	2.5±1.5	A, <.04
IS_{OGTT}	6.5±2.9	3.9±2.1	6.7±4.3	5.8±3.4	A, <.01 B, <.035
CIRgp	1.4±0.9	1.2±0.88	1.3±0.0.96	1.11±0.97	NS

A, main effect of treatment for both; B, interaction effect — RING versus PILL posttreatment; NS, not significant.
The HOMA index is based on fasting basal values of insulin and glucose and is a better representative of the metabolic state because it controls for the levels of insulin relative to the glucose level. A single fasting insulin or glucose value is too variable between subjects and within a subject, and therefore, it is difficult to find meaningful differences due to treatment.

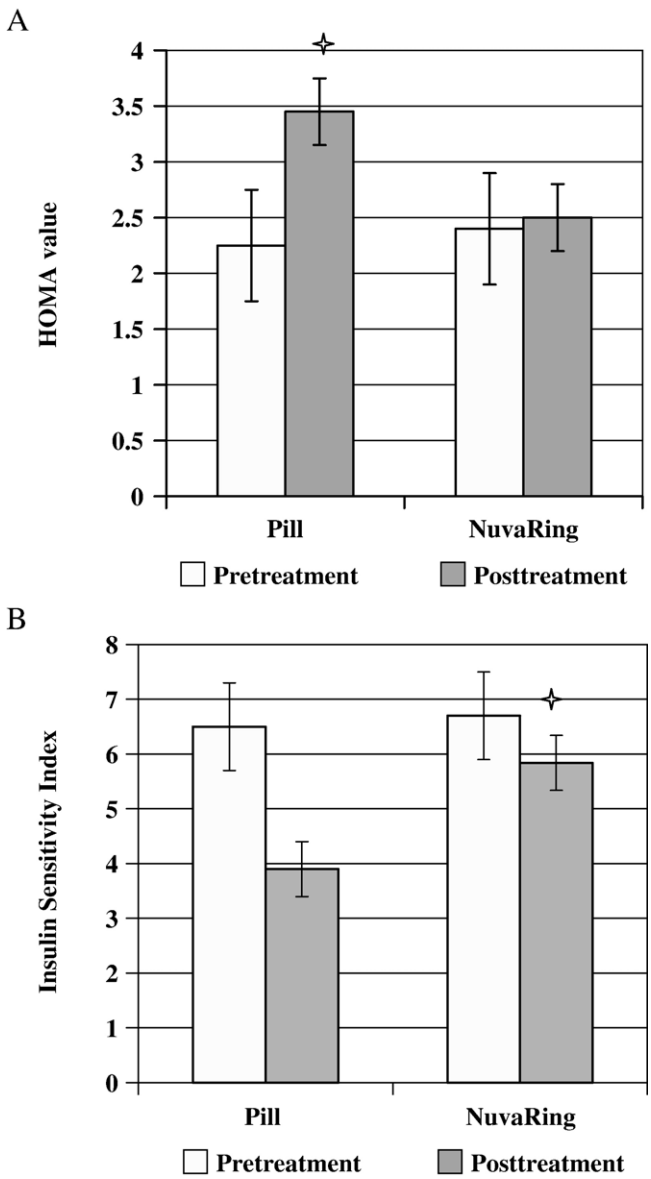

Fig. 2. Insulin sensitivity measurements at baseline and after 5 months of contraceptive treatment in the PILL and RING groups (only patients who completed the study are included). Data shown are the mean±SE for basal fasting insulin resistance (HOMA) index (A) (*p<.04, pretreatment vs. during treatment for PILL) and insulin sensitivity index (IS_{OGTT}) derived from a postglucose challenge (B) (*p<.035, PILL vs. RING during contraceptive treatment).

treatment in the RING group (p<.001) but not in the PILL group. Mean fasting plasma cholesterol, triglycerides, HDL cholesterol and calculated LDL cholesterol concentrations did not change consistently when compared with baseline values with either treatment (Table 1).

3.3. Tolerability

Tolerability of both contraceptives was good, and AEs were comparable for the two treatment groups. The number and proportion of subjects in each group who reported AEs that were defined by the investigators as "definitely" related to treatment are shown in Table 2. A slightly higher proportion of subjects in the PILL group experienced an AE that was considered to be drug related by investigators: 17% in the RING group and 29.0% in the PILL group. The most frequently reported AEs were irregular breakthrough bleeding for the PILL group (17%) and vaginal symptoms (9.7%) in the RING group. Patients desiring to continue

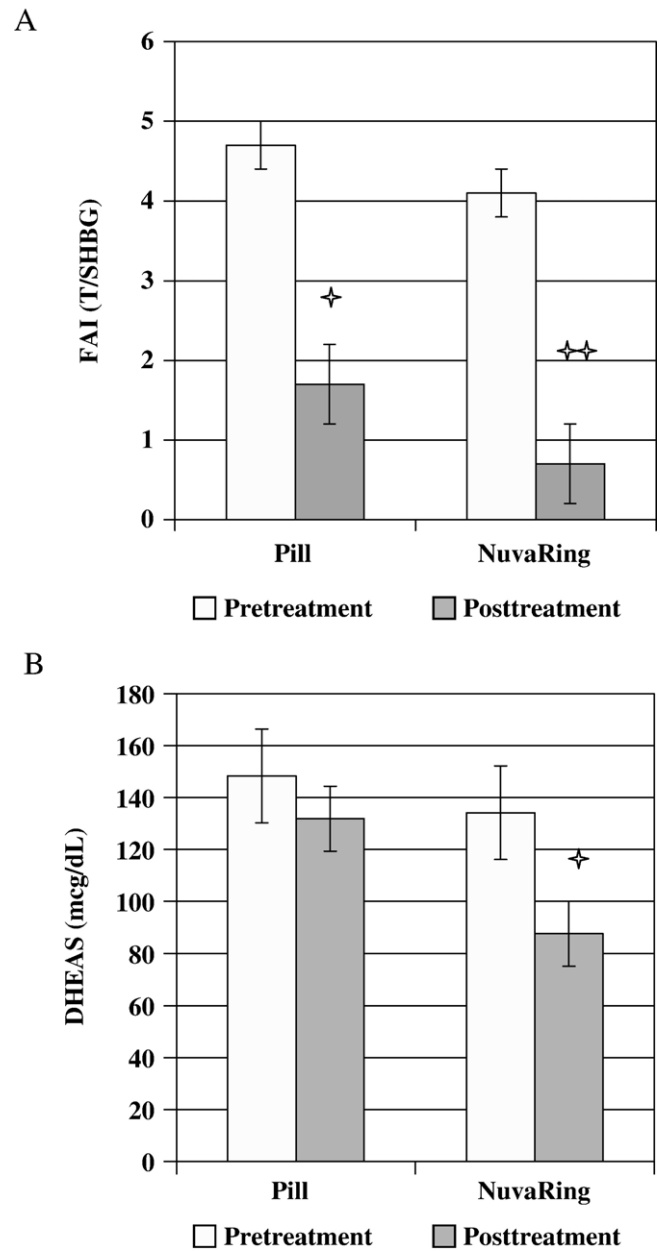

Fig. 3. Baseline and posttreatment in the PILL and RING groups for FAI measurements (A) and adrenal androgen (DHEAS) levels (B). Data are presented as mean±SE for FAI (*p<.003, pretreatment vs. during treatment for PILL treatment, and **p<.0005, pretreatment vs. during treatment for RING treatment) and mean±SE for DHEAS (*p<.01, PILL vs. RING during contraceptive treatment).

Table 2
Proportion of subjects who reported treatment-related AEs (intent-to-treat population)

Description	PILL (n=24)	RING (n=31)
Breakthrough bleeding	4 (17)	0 (0)
Breast tenderness	1 (4)	1 (3)
Hot flashes	1 (4)	0 (0)
Mood swings	1 (4)	1 (3)
Vaginal yeast infection/discomfort	0 (0)	3 (9.7)

Values are expressed as n (%).

NuvaRing® or OC pill use were provided an extra month of medication. Patients using contraceptive pills were offered the opportunity to use the NuvaRing® for 1 month at the end of the trial.

4. Discussion

Limited evidence raises the issue that OC therapy may alter insulin resistance and exert other untoward metabolic actions that possibly enhance the long-term risk for diabetes and heart disease. In this study, we investigated the impact of oral versus vaginal hormonal contraceptive therapy on insulin sensitivity and secretion in women of reproductive age. Our results showed that use of a low-dose combination OC reduced insulin sensitivity to a larger extent than the contraceptive vaginal ring. We observed an effect on carbohydrate metabolism with oral intake as evidenced by higher insulin levels after a glucose challenge, which was not evident with vaginal administration. This effect on insulin in the PILL group occurred despite a stable BMI and marked reduction in circulating androgens. While no participants presented with impaired fasting glucose or IGT prior to contraceptive therapy, two normal cycling women on PILL therapy developed IGT after 5 months of treatment, which was not found with any subjects in the RING group. These findings suggest that healthy reproductive-age women who are considering initiating oral combined contraceptive therapy should consider that their insulin sensitivity might be altered with treatment. Given that OCs may deteriorate insulin sensitivity, their use may aggravate the already impaired glucose metabolism of obese women and women with PCOS or metabolic syndrome. Thus, obese women and/or those with glucose intolerance at baseline may be at a higher risk for the adverse effects from OCs. From a metabolic standpoint, vaginal hormonal contraceptive therapy would appear to be preferable for these women.

Inconsistencies in the results of studies of OCs and carbohydrate metabolism might also reflect assessment of different formulations. The new progestins developed for contraceptives have been designed to be less androgenic and more like the natural hormone progesterone [27]. In addition, the metabolic effects may vary with route of administration of hormonal contraception. OC pills affect carbohydrate metabolism, dose-dependently, as do androgens and progestins of greater androgenicity. Vaginal administration of contraceptive hormones allows lower dosing than oral pills because hormones are delivered systemically [20]. Given that the synthetic progestins used in OCs are associated with a dose-dependent impairment of carbohydrate metabolism, the lesser effect of the vaginal ring on insulin resistance may be due to its controlled steady-state release of hormones and lower systemic hormone dose exposure [20,22].

We compared the influence of the RING with PILL treatment on ovarian hormones and lipid levels. Both hormonal contraceptives were found to suppress ovarian androgen production, increase SHBG levels and reduce free T. It has been previously shown that vaginal administration of EE increases SHBG serum concentrations as well as oral EE, underscoring (perhaps counterintuitively) the fact that parenteral as well as oral EE administration can increase hepatic protein synthesis [28]. Additionally, studies found that OCs containing EE generally increase the synthesis of SHBG, but the progestin component modifies this effect [29]. Perhaps because the OC formulation used in this study combines a low dose of estrogen with the androgenic progestin levonorgestrel, the associated increase in SHBG is modest. Etonogestrel released by the vaginal ring is associated with a higher SHBG increase, as it opposes the estrogenic effect to a much lower extent [29]. While both contraceptives lowered T, there was a significantly greater reduction of free androgen in the RING users. Since many women seek treatment for the manifestations of hyperandrogenism, such as hirsutism and acne, the greater stimulation of SHBG production with RING suggests that vaginal contraceptive administration may be the more appropriate treatment for both women with hirsutism or PCOS and normal women prone to androgenic side effects.

An intriguing finding in our study was a significant decrease in DHEAS in the RING group, which was not observed in patients in the PILL group. Whether this effect was due to a steady-state suppression of vaginal administration is unclear but plausible. Importantly, while both hormonal contraceptives tested altered lipid levels to some extent, there were no statistically significant effects on lipid metabolism with either contraceptive in this trial. This is consistent with previous studies, which demonstrated that the dose and type of progestin in contraceptives determine the effects of OCs on many metabolic markers [12,30]. Importantly, while all combined OCs alter lipid levels to some extent, there were no statistically significant differences in lipid effects among OC preparations.

The occurrence of AEs and use of concomitant medication were recorded throughout the trial. The incidence of treatment-related complaints was low for both contraceptives. Most women found the ring easy to use. RING users in contrast to the PILL users reported no complaints of breakthrough bleeding, consistent with other reports [31]. The very low incidence of breakthrough bleeding with the

vaginal ring is in all probability related to the continuous and steady release of the contraceptive steroids [31]. Specific vaginal-ring-related events, such as vaginal discomfort, were also captured as AEs and reported by 9.7% of the RING users. Breast pain (patients self-reported during office visits or phone follow-ups — no questionnaire concerning side effects was used) occurred in 4% of women in the PILL group and 3% in the RING group. As this complaint is likely due to the estrogen content of the hormonal contraceptive preparation, it is likely that the low incidence of breast pain in this study is related to the low dose of EE with both the RING and the PILL.

In choosing hormonal contraceptives, identifying which formulation causes the least insulin resistance may be important. While a reduction in insulin action in OC users was identified by both fasting (HOMA) and glucose-stimulated (IS_{OGTT}) insulin sensitivity indices used in this study, women in the PILL group showed only modest changes in basal glucose and insulin levels. However, these subjects demonstrated significant increases in glucose and insulin levels after a glucose challenge compared to women in the RING group. The HOMA index, based on a single fasting glucose and insulin value, may be more reflective of changes in hepatic insulin sensitivity and generally provides a weaker estimate of total body insulin sensitivity. This would lead to limitations of HOMA in detecting changes in peripheral insulin sensitivity with hormonal contraception. Dynamic testing appears to be necessary to better demonstrate the diminished insulin responsiveness with OC use. We therefore suggest that, although quite adequate in other contexts, the HOMA model provides an inadequate assessment of insulin sensitivity when assessing the impact of hormonal contraceptive therapy. This is consistent with other reports that show that glucose and insulin calculations that involve fasting values do not correlate with insulin sensitivity as measured by the gold-standard method, the euglycemic clamp [32].

Our study confirms that even low-dose contraceptive pills commonly prescribed in healthy reproductive-age women had measurable impact on metabolic function. The contraceptive vaginal ring had less impact on carbohydrate metabolism. Low-dose OCs are generally prescribed without concern about precipitation of hyperglycemia, and clinical monitoring of glucose levels is not recommended in healthy women. There is little evidence of the best practice for the follow-up of women at risk for diabetes prescribed by hormonal contraception. These findings provide important information to be used when counseling patients regarding contraceptive methods, particularly women at risk for metabolic disorders such as previous gestational diabetes, obesity or ovarian hyperandrogenism.

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